

Chapter 5

The Emitter Follower

(Common Collector Stage)

Voltage Buffering, Level Shifting, High R_{in} , Low R_{out}

*Based on Razavi, Fundamentals of Microelectronics, 2nd Ed.
Chapter 5.3.3 (Emitter Follower)*

With insights from Razavi's BJT YouTube Lectures (27–28)

Topics Covered:

- Physical intuition: why the emitter “follows” the base
- Why the voltage gain is always less than 1 (and why that’s useful!)
- Full small-signal derivation of voltage gain
- Input impedance: very high ($r_{\pi} + (\beta + 1)R_E$)
- Output impedance: very low ($1/g_m$)
- The Thevenin equivalent perspective (voltage divider model)
- Follower with source resistance
- Follower with ideal current source ($A_v = 1$)
- The role of the follower as a voltage buffer
- Complete comparison of all three topologies (CE, CB, CC)
- Practice problems and common mistakes

Chapter 5: Quick Reference Formula Sheet

Emitter Follower Core Parameters

Parameter	Formula	Notes
Voltage gain	$A_v = \frac{R_E}{R_E + 1/g_m}$	Positive, always < 1
With current source	$A_v = 1$	$R_E \rightarrow \infty$
Input impedance	$R_{in} = r_\pi + (\beta + 1)R_E$	Very high (impedance reflection)
Output impedance	$R_{out} = \frac{1}{g_m}$	Very low ($\sim 26 \Omega$ at 1 mA)
Level shift	$V_{out} = V_{in} - V_{BE}$	≈ 0.7 V drop

Follower with Source Resistance

Parameter	Formula	Notes
Voltage gain	$A_v = \frac{R_E}{R_E + \frac{R_S}{\beta+1} + \frac{1}{g_m}}$	R_S divided by $\beta + 1$
Output impedance	$R_{out} = \frac{1}{g_m} + \frac{R_S}{\beta + 1}$	Source resistance reflected

Thevenin Equivalent Model

Parameter	Formula	Notes
Thevenin voltage	$v_{Thev} = v_{in}$	Open-circuit: $R_E = \infty \implies A_v = 1$
Thevenin resistance	$R_{Thev} = 1/g_m$	Looking back into the emitter

1 The Emitter Follower: Physical Intuition

1.1 What Does “Follower” Mean?

The **emitter follower** (also called the “common-collector stage”) is the third and final BJT amplifier topology. Its defining property is that the output voltage at the emitter **follows** the input voltage at the base, shifted down by $V_{BE} \approx 0.7$ V.

we first study the core and subsequently add the biasing elements.

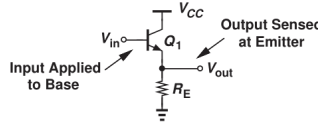


Figure 1: Figure 5.83 from Razavi — The emitter follower. Input is at the base, output is at the emitter. The collector is tied to V_{CC} (AC ground).

Physical intuition — step by step:

1. Suppose V_{in} increases by a small ΔV_{in} .
2. V_{BE} tends to increase, which increases I_C (and I_E).
3. The higher I_E flowing through R_E increases $V_{out} = I_E R_E$.
4. But as V_{out} rises, $V_{BE} = V_{in} - V_{out}$ is *partially restored*, limiting how much I_E increases.
5. **Key insight:** V_{out} *cannot* rise more than V_{in} . If it did, V_{BE} would decrease, reducing I_E , which would pull V_{out} back down. This is negative feedback.
6. **Result:** $\Delta V_{out} < \Delta V_{in}$, so the voltage gain is **positive but less than 1**.

■ KEY RESULT:

The emitter follower has a voltage gain **less than unity** ($0 < A_v < 1$). The emitter voltage “follows” the base voltage with approximately a constant V_{BE} drop. In the DC domain: $V_{out} = V_{in} - V_{BE}$. In small-signal: $v_{out} \approx v_{in}$ (for $g_m R_E \gg 1$).

CONCEPT: Why is a Gain < 1 Useful?

At first glance, an amplifier with gain less than 1 seems useless. But the follower provides something the CE and CB stages cannot:

- **Very high input impedance** $\sim (\beta + 1)R_E$ — draws negligible current from the source.
- **Very low output impedance** $\sim 1/g_m$ — can drive heavy loads without gain loss.
- **No phase inversion** — the output tracks the input exactly.

The follower is a **voltage buffer**: it doesn’t amplify voltage, but it transforms impedance levels. It takes a “weak” high-impedance signal and makes it capable of driving low-impedance loads.

LECTURE INSIGHT (*Lecture 27*)

Razavi explains: “The emitter follower doesn’t amplify voltage — it amplifies *current*. It takes a signal from a source that can’t deliver much current and reproduces it at a node that *can*. This impedance transformation is one of the most valuable functions in analog design. Almost every real amplifier has a follower somewhere in its chain.”

2 Large-Signal and I/O Behavior

LEVEL SHIFT.

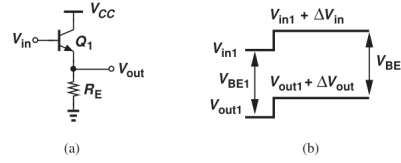


Figure 5.84 (a) Emitter follower sensing an input change. (b) Response of the circuit.

Figure 2: Figure 5.84 from Razavi — (a) Emitter follower sensing an input change. When V_{in} rises by ΔV , V_{out} also rises, but by slightly less than ΔV . (b) The I/O characteristic: slope is close to but less than 1.

The I/O relationship is:

$$V_{out} = V_{in} - V_{BE}$$

Since $V_{BE} = V_T \ln(I_C/I_S)$ and $I_C = V_{out}/R_E$, there is a weak logarithmic dependence. For small signals, the slope of the I/O curve is:

$$\frac{\partial V_{out}}{\partial V_{in}} = \frac{R_E}{R_E + 1/g_m} < 1$$

Why can ΔV_{out} never exceed ΔV_{in} ? Razavi gives an elegant proof by contradiction: Suppose $\Delta V_{out} > \Delta V_{in}$. Then $V_{BE, \text{new}} = V_{in, \text{new}} - V_{out, \text{new}} < V_{BE, \text{old}}$. But a *smaller* V_{BE} means *less* I_E , which means a *smaller* $V_{out} = I_E R_E$ — contradicting the assumption. Therefore $\Delta V_{out} \leq \Delta V_{in}$, and $A_v \leq 1$.

3 Small-Signal Voltage Gain

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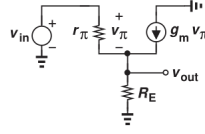


Figure 5.85 Small-signal model of emitter follower

Figure 3: Figure 5.85 from Razavi — Small-signal equivalent circuit of the emitter follower. Note that V_{CC} is AC ground, so the collector is grounded.

DERIVATION: Emitter Follower Voltage Gain

From the small-signal model ($V_A = \infty$, collector grounded):

Step 1: KCL at the emitter node — the current through R_E equals the sum of v_π/r_π and $g_m v_\pi$:

$$\frac{v_{out}}{R_E} = \frac{v_\pi}{r_\pi} + g_m v_\pi = v_\pi \left(\frac{1}{r_\pi} + g_m \right)$$

Solving for v_π :

$$v_\pi = \frac{v_{out}}{R_E \left(\frac{1}{r_\pi} + g_m \right)} = \frac{r_\pi}{\beta + 1} \cdot \frac{v_{out}}{R_E}$$

Step 2: KVL from input to output:

$$v_{in} = v_\pi + v_{out}$$

Step 3: Substitute v_π from Step 1:

$$v_{in} = \frac{r_\pi}{(\beta + 1)R_E} v_{out} + v_{out} = v_{out} \left(1 + \frac{r_\pi}{(\beta + 1)R_E} \right)$$

Since $r_\pi/(\beta + 1) = 1/g_m$:

$$v_{in} = v_{out} \left(1 + \frac{1}{g_m R_E} \right) = v_{out} \cdot \frac{R_E + 1/g_m}{R_E}$$

$$A_v = \frac{v_{out}}{v_{in}} = \frac{R_E}{R_E + 1/g_m}$$

The gain is always positive and always less than 1.

Important Limiting Cases

Emitter Follower Gain Limits

Case	Condition	Gain
Generic	Any R_E	$A_v = R_E/(R_E + 1/g_m)$
Large R_E	$R_E \gg 1/g_m$	$A_v \approx 1$
Current source load	$R_E \rightarrow \infty$	$A_v = 1$ (exactly!)
Small R_E	$R_E \ll 1/g_m$	$A_v \approx g_m R_E \ll 1$

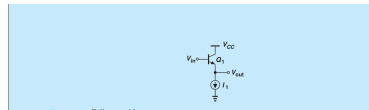


Figure 4: Figure 5.86 from Razavi — Emitter follower with an ideal current source.

Example 5.43: Follower with Ideal Current Source (Razavi)

Problem: In integrated circuits, the follower typically uses a current source instead of R_E (Fig. 5.86). What is the gain if the current source is ideal and $V_A = \infty$?

Solution:

An ideal current source has infinite impedance. Setting $R_E \rightarrow \infty$ in the gain formula:

$$A_v = \frac{R_E}{R_E + 1/g_m} \rightarrow 1$$

Alternatively: if I_C is constant (forced by the current source), then $V_{BE} = V_T \ln(I_C/I_S)$ is constant. Since $V_{out} = V_{in} - V_{BE}$ and V_{BE} doesn't change, $\Delta V_{out} = \Delta V_{in}$ exactly.

This is why current-source-loaded followers are preferred in ICs: they achieve unity gain.

EXAM TIP

Quick exam formula: The follower gain is always a voltage divider between R_E and $1/g_m$. Think of $1/g_m$ as the “internal resistance” of the emitter follower. The output gets $R_E/(R_E + 1/g_m)$ of the input, just like a simple resistor divider.

4 Input Impedance

The emitter follower has the highest input impedance of all three topologies.

DERIVATION: Emitter Follower Input Impedance

Looking into the base, the impedance reflection rule from Chapter 3 applies directly. Any impedance R_E at the emitter appears multiplied by $(\beta + 1)$ at the base:

$$R_{in} = r_\pi + (\beta + 1)R_E$$

For $\beta = 100$ and $R_E = 1 \text{ k}\Omega$: $R_{in} = r_\pi + 101 \times 1\text{k} \approx 101 \text{ k}\Omega + r_\pi \approx \mathbf{103.6 \text{ k}\Omega}$

With a current source load ($R_E \rightarrow \infty$): $R_{in} \rightarrow \infty$ (the base draws zero small-signal current!).

CONCEPT: Why the Follower Has Such High R_{in}

Physical explanation: When a test voltage v_X is applied to the base, the emitter follows nearly identically. So the voltage across r_π (which is $v_{BE} = v_X - v_E$) is very small. Since $i_B = v_{BE}/r_\pi$ and $v_{BE} \approx 0$, the base current is tiny. A tiny current for a given voltage means a very high impedance.

This is the essence of negative feedback: the follower's output "opposes" the input, making the transistor appear nearly transparent. Almost no current is drawn from the source.

LECTURE INSIGHT (Lecture 28)

Razavi notes: "The emitter follower's high input impedance comes from negative feedback. The emitter tries to follow the base, which means V_{BE} hardly changes, which means the base current hardly changes. From the source's perspective, the transistor looks like it's not even there — it just passes the voltage through while drawing almost no current."

5 Output Impedance

The follower's output impedance is as notable as its input impedance — but in the opposite direction: it is remarkably **low**.

DERIVATION: Emitter Follower Output Impedance

To find R_{out} , set $v_{in} = 0$ and apply a test voltage v_X at the emitter. With the base grounded:

$$v_{\pi} = 0 - v_X = -v_X$$

KCL at the emitter gives:

$$i_X = \frac{v_X}{R_E} + \frac{-v_{\pi}}{r_{\pi}} + g_m(-v_{\pi}) = \frac{v_X}{R_E} + v_X \left(\frac{1}{r_{\pi}} + g_m \right)$$

The output impedance from the emitter node is:

$$R_{out} = \frac{v_X}{i_X} = R_E \parallel \frac{1}{g_m + 1/r_{\pi}} \approx R_E \parallel \frac{1}{g_m}$$

Since $R_E \gg 1/g_m$:

$$R_{out} \approx \frac{1}{g_m}$$

This is exactly the same $1/g_m$ we saw in the CB stage's input impedance! This is not a coincidence: in both cases, we are looking into the emitter of a transistor with the base at AC ground.

5.1 Output Impedance with Source Resistance

If the signal source driving the base has resistance R_S , the output impedance increases:

$$R_{out} = \frac{1}{g_m} + \frac{R_S}{\beta + 1}$$

The source resistance R_S is reflected from the base to the emitter, **divided by** $(\beta + 1)$ — the same inverse impedance reflection rule from the CB stage (Chapter 4).

EXAM TIP

For exams: The follower's output impedance formula $R_{out} = 1/g_m + R_S/(\beta + 1)$ is the **exact mirror** of its input impedance $R_{in} = r_{\pi} + (\beta + 1)R_E$:

- Looking from base $\rightarrow$ emitter: impedances are **multiplied** by $(\beta + 1)$.
- Looking from emitter $\rightarrow$ base: impedances are **divided** by $(\beta + 1)$.

6 The Thevenin Equivalent: Voltage Divider Model

that shown in Fig. 5.87(a), combining operation as a voltage divider.

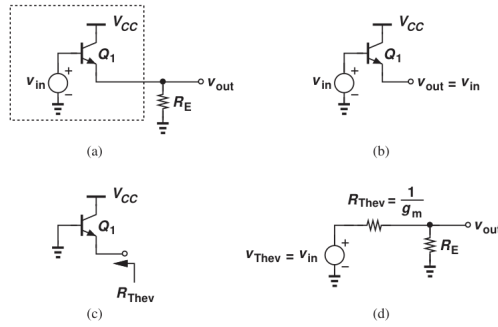


Figure 5.87 (a) Emitter follower stage. (b) Thevenin voltage. (c) Thevenin resistance.

Figure 5: Figure 5.87 from Razavi — (a) Emitter follower stage. (b) Open-circuit output: $v_{Th} = v_{in}$ (with $R_E = \infty$, $A_v = 1$). (c) Thevenin resistance: $R_{Th} = 1/g_m$. (d) Simplified equivalent: a voltage divider between $1/g_m$ and R_E .

A powerful way to understand the follower is through its **Thevenin equivalent**:

DERIVATION: Thevenin Model of the Follower

Thevenin voltage: Open the output ($R_E = \infty$). From Example 5.43, we already know that the follower with $R_E = \infty$ has $A_v = 1$. Therefore:

$$v_{Th} = v_{in}$$

Thevenin resistance: Set $v_{in} = 0$ (base grounded) and look into the emitter. We already derived this: $R_{Th} = 1/g_m$.

Equivalent circuit: The follower reduces to v_{in} in series with $1/g_m$, driving R_E :

$$v_{out} = v_{in} \cdot \frac{R_E}{R_E + 1/g_m}$$

This is simply a **resistive voltage divider** between $1/g_m$ (series resistance) and R_E (load).

KEY RESULT:

The Thevenin model confirms that the follower acts as a **near-ideal voltage source** with output resistance $1/g_m$. The signal passes through with almost no loss (since $1/g_m$ is so small), making the follower the transistor equivalent of a “wire with gain = 1 and very low output impedance.”

Example 5.44: Follower with Source Impedance (Razavi)

Problem: A follower is driven by a source v_{in} with impedance R_S . Find the voltage gain.

Solution:

Using the Thevenin approach: model v_{in} , R_S , and Q_1 as a single Thevenin equivalent.

- The open-circuit output voltage is still v_{in} (unloaded follower gain = 1).

- The Thevenin resistance is $R_S/(\beta + 1) + 1/g_m$ (using the CB input impedance formula with $R_B = R_S$).

The equivalent circuit is a divider:

$$A_v = \frac{R_E}{R_E + \frac{R_S}{\beta+1} + \frac{1}{g_m}}$$

For $R_S = 10 \text{ k}\Omega$, $\beta = 100$, $g_m = 38.5 \text{ mA/V}$, $R_E = 1 \text{ k}\Omega$:

$$A_v = \frac{1000}{1000 + 99 + 26} = \frac{1000}{1125} = \mathbf{0.89}$$

Compare: without R_S , $A_v = 1000/(1000 + 26) = 0.975$. The source resistance degrades the gain, but only mildly because R_S is divided by $(\beta + 1)$.

7 The Emitter Follower as a Voltage Buffer

7.1 The Problem It Solves

Recall from Chapter 3 that connecting a low-impedance load (like a speaker with $R_{sp} = 8\ \Omega$) directly to a CE amplifier's output destroyed the gain: $A_v = -g_m(R_C \parallel R_{sp}) \approx -g_m \times 8 \approx 0.3$. The solution is to insert a **follower** between the CE stage and the load.

CONCEPT: The Buffer Chain: CE + Follower

A practical amplifier chain:

1. **CE stage:** Provides high voltage gain $|A_v| = g_m R_C \approx 40$.
2. **Emitter follower:** Provides $A_v \approx 1$ but transforms the high output impedance of the CE ($R_C \approx 1\ \text{k}\Omega$) into low output impedance ($1/g_m \approx 26\ \Omega$).
3. **Load (8 Ω speaker):** Now sees $R_{out} \approx 26\ \Omega$, so the voltage divider loss is only $8/(8 + 26) \approx 24\%$ instead of 99%.

Overall gain: $\approx 40 \times 1 \times 0.76 \approx 30$. Without the follower: ≈ 0.3 .

The follower increased the usable gain by 100 $\times$!

7.2 Summary of Buffer Properties

Emitter Follower as Voltage Buffer

An ideal voltage buffer has:

- $A_v = 1$ (no voltage loss) ✓ (follower: $A_v \approx 1$ for large R_E)
- $R_{in} = \infty$ (draws no current from source) ✓ (follower: $R_{in} = r_\pi + (\beta + 1)R_E \gg R_S$)
- $R_{out} = 0$ (drives any load) ✓ (follower: $R_{out} = 1/g_m \approx 26\ \Omega$)

The emitter follower approximates all three properties of an ideal buffer. It is the closest thing to a “perfect wire” that a single transistor can provide.

8 Complete Comparison: CE vs. CB vs. CC

The Three BJT Amplifier Topologies — Complete Summary

Property	CE (Ch. 3)	CB (Ch. 4)	CC (Ch. 5)
Input	Base	Emitter	Base
Output	Collector	Collector	Emitter
Common	Emitter	Base	Collector
A_v	$-g_m R_C$	$+g_m R_C$	$\frac{R_E}{R_E + 1/g_m}$
Sign	Inverting	Non-inverting	Non-inverting
$ A_v $	Can be $\gg 1$	Can be $\gg 1$	Always < 1
R_{in}	r_π (moderate)	$1/g_m$ (low)	$r_\pi + (\beta + 1)R_E$ (high)
R_{out}	R_C (moderate)	R_C (moderate)	$1/g_m$ (low)
A_I	β	≈ 1	$\beta + 1$
Best for	Voltage gain	RF, cascode	Buffer, driver

CONCEPT: How the Three Topologies Complement Each Other

Each topology excels in a different dimension:

- **CE:** The “workhorse” for voltage gain. Use when you need to amplify a signal.
- **CB:** The “speed demon” for high-frequency circuits. Use when bandwidth matters (RF, cascodes).
- **CC (Follower):** The “impedance transformer.” Use when you need to drive a load without losing signal.

A complete amplifier system often uses all three: a CE stage for gain, a CB stage for bandwidth (as part of a cascode), and a CC stage as an output buffer.

LECTURE INSIGHT (Lecture 28)

Razavi concludes the amplifier topology series: “Every amplifier you will ever see in practice is built from combinations of these three fundamental topologies. Once you understand CE, CB, and CC — their gains, impedances, and trade-offs — you can analyze any circuit, no matter how complex. This is the foundation of all analog design.”

9 Practice Problems

Problem 1: Basic Follower Analysis

An emitter follower has $I_C = 1$ mA, $R_E = 500\ \Omega$, $\beta = 100$, $V_A = \infty$.

- Find g_m , A_v , R_{in} , and R_{out} .
- By what factor is R_{in} larger than the CE stage's R_{in} (with same bias)?
- By what factor is R_{out} smaller than the CE stage's R_{out} ($R_C = 1$ k Ω)?

Solution:

(a) $g_m = 1\text{m}/26\text{m} = \mathbf{38.5}$ mA/V. $1/g_m = 26\ \Omega$. $r_\pi = 100/38.5\text{m} = 2.6$ k Ω .

$A_v = R_E/(R_E + 1/g_m) = 500/(500 + 26) = \mathbf{0.95}$.

$R_{in} = r_\pi + (\beta + 1)R_E = 2600 + 101 \times 500 = 2600 + 50500 = \mathbf{53.1}$ k Ω .

$R_{out} = 1/g_m = \mathbf{26\ \Omega}$.

(b) CE $R_{in} = r_\pi = 2.6$ k Ω . Follower/CE = $53.1\text{k}/2.6\text{k} = \mathbf{20.4\times}$.

(c) CE $R_{out} = R_C = 1$ k Ω . CE/Follower = $1000/26 = \mathbf{38.5\times}$.

Problem 2: Follower with Source Resistance

A follower has $I_C = 2$ mA, $R_E = 1$ k Ω , $\beta = 150$, $R_S = 50$ k Ω .

- Find the gain and the output impedance.
- What gain would the follower have without R_S ?
- A $50\ \Omega$ load is connected. What is the voltage at the load?

Solution:

$g_m = 2\text{m}/26\text{m} = 76.9$ mA/V. $1/g_m = 13\ \Omega$.

(a) $A_v = R_E/(R_E + R_S/(\beta + 1) + 1/g_m) = 1000/(1000 + 50000/151 + 13) = 1000/(1000 + 331 + 13) = 1000/1344 = \mathbf{0.744}$.

$R_{out} = 1/g_m + R_S/(\beta + 1) = 13 + 331 = \mathbf{344\ \Omega}$.

(b) Without R_S : $A_v = 1000/(1000 + 13) = \mathbf{0.987}$. The source resistance degrades gain significantly.

(c) With $50\ \Omega$ load: $v_{load} = v_{out} \times 50/(50 + 344) = 0.744v_{in} \times 0.127 = \mathbf{0.094\ v_{in}}$. The high R_{out} (due to large R_S) makes this a poor driver!

Problem 3: Follower as Buffer Design

Design a follower to buffer a CE stage output ($R_{out,CE} = 2$ k Ω , $A_{v,CE} = -30$) so that it can drive an $8\ \Omega$ speaker. The follower uses $I_C = 5$ mA, $\beta = 100$.

- Find the follower's R_{out} .
- Find the overall gain (CE + follower + speaker).
- Compare with directly connecting the speaker to the CE stage.

Solution:

$g_m = 5\text{m}/26\text{m} = 192$ mA/V. $1/g_m = 5.2\ \Omega$.

(a) $R_{out} = 1/g_m + R_S/(\beta + 1) = 5.2 + 2000/101 = 5.2 + 19.8 = \mathbf{25\ \Omega}$.

(b) The speaker sees a divider: $v_{speaker}/v_{follower\ input} = 8/(8 + 25) = 0.242$.

Follower gain (with $R_E = 8\ \Omega$ speaker as load): $A_{v,follower} = R_E/(R_E + 1/g_m + R_S/(\beta + 1)) = 8/(8 + 25) = 0.242$.

Overall: $|A_v| = 30 \times 0.242 = \mathbf{7.3}$.

(c) Without follower: $A_v = -g_{m,CE}(R_C || 8) \approx -g_m \times 8 \approx \mathbf{0.3}$. The follower improves the usable gain by $\mathbf{24\times}$!

Problem 4: Current Source Follower

A follower uses an ideal current source (instead of R_E) with $I_C = 0.5 \text{ mA}$, $\beta = 200$, $R_S = 10 \text{ k}\Omega$.

- (a) Find the voltage gain.
- (b) Find R_{in} and R_{out} .

Solution:

$$g_m = 0.5\text{mA}/26\text{mV} = 19.2 \text{ mA/V}. \quad r_\pi = 200/19.2\text{mA/V} = 10.4 \text{ k}\Omega.$$

$$(a) \quad R_E \rightarrow \infty: A_v = R_E / (R_E + R_S / (\beta + 1) + 1/g_m) \rightarrow 1 / (R_S / ((\beta + 1)R_E) + 1/g_m R_E + 1).$$

As $R_E \rightarrow \infty$: the R_S and $1/g_m$ terms vanish in the denominator. $A_v = 1$.

$$(b) \quad R_{in} = r_\pi + (\beta + 1) \times \infty \rightarrow \infty \text{ (draws zero small-signal current).}$$

$$R_{out} = 1/g_m + R_S / (\beta + 1) = 52 + 10000/201 = 52 + 49.8 = \mathbf{102 \, \Omega}.$$

10 Common Mistakes and How to Avoid Them

COMMON MISTAKES

- **Mistake:** Saying the follower has “no gain” or is “useless.”

Fix: The follower provides **impedance transformation**, which is invaluable. It converts high-impedance signals into low-impedance ones. It also provides current gain $\approx \beta + 1$.
- **Mistake:** Writing $A_v = 1$ without checking R_E .

Fix: $A_v = 1$ only if $R_E \rightarrow \infty$ (ideal current source). With finite R_E : $A_v = R_E / (R_E + 1/g_m) < 1$. For $R_E = 100 \, \Omega$ and $1/g_m = 26 \, \Omega$: $A_v = 100/126 = 0.79$ (not even close to 1!).
- **Mistake:** Confusing R_{in} of the follower with R_{in} of the CE stage.

Fix: CE: $R_{in} = r_\pi$. Follower: $R_{in} = r_\pi + (\beta + 1)R_E$ — much larger because R_E is reflected through $(\beta + 1)$.
- **Mistake:** Writing $R_{out} = R_E$ for the follower.

Fix: $R_{out} = 1/g_m \parallel R_E \approx 1/g_m$ (since $1/g_m \ll R_E$). The transistor’s low emitter impedance dominates. R_E is seen in *parallel*, not series!
- **Mistake:** Forgetting $R_S/(\beta + 1)$ in the output impedance.

Fix: If the source has impedance R_S , it adds $R_S/(\beta + 1)$ to R_{out} . This can be significant if R_S is large: $R_{out} = 1/g_m + R_S/(\beta + 1)$.
- **Mistake:** Claiming the follower inverts the signal.

Fix: The follower is **non-inverting**. $A_v > 0$ always. The output is a faithful (but slightly attenuated) copy of the input.
- **Mistake:** Thinking the follower is the same as a CE stage with $R_C = 0$.

Fix: Setting $R_C = 0$ in a CE stage gives $A_v = -g_m \times 0 = 0$ (no gain at all!). The follower and the CE stage have fundamentally different signal paths: in the follower, the output is at the emitter, not the collector.

Chapter 5: End-of-Chapter Summary

Core Emitter Follower

- **Topology:** Input at base, output at emitter, collector to V_{CC} (AC ground).
- **Gain:** $A_v = R_E / (R_E + 1/g_m)$. Always positive, always < 1 .
- **With current source:** $A_v = 1$ (perfect follower).
- **Level shift:** $V_{out} = V_{in} - V_{BE}$ (constant ≈ 0.7 V drop).

Impedances

- **Input impedance:** $R_{in} = r_\pi + (\beta + 1)R_E$ — very high. R_E is reflected up by $(\beta + 1)$.
- **Output impedance:** $R_{out} = 1/g_m$ — very low ($\sim 26 \Omega$).
- **With source resistance:** $R_{out} = 1/g_m + R_S/(\beta + 1)$ — R_S is reflected down.

Thevenin Model

- $v_{Thev} = v_{in}$, $R_{Thev} = 1/g_m$.
- The follower is a voltage divider between $1/g_m$ (series) and R_E (load).

Role as Buffer

- Approximates an ideal voltage buffer: $A_v \approx 1$, $R_{in} \rightarrow \infty$, $R_{out} \rightarrow 0$.
- Essential when a high-gain stage (CE) must drive a low-impedance load.
- The cascade CE $\rightarrow$ Follower is one of the most common structures in analog design.

Key Design Trade-offs

Choice	Benefit	Cost
Larger R_E	Gain closer to 1	More voltage headroom
Current source load	$A_v = 1$ exactly	More circuit complexity
Higher I_C	Lower R_{out}	More power dissipation
Lower R_S	Better gain, lower R_{out}	May not be controllable

“The emitter follower is the unsung hero of analog design. It doesn’t amplify, but without it, no amplifier could drive a real load. Every great amplifier has a follower at the end.” — Razavi